

Manish Kajla

Second Year Undergraduate
Department of Mathematics and Statistics

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2024-Present	B.S.	Indian Institute of Technology Kanpur	5.7/10
2023	RBSE(XII)	Prince School Sikar	90%
2021	RBSE(X)	Shri Dhruva Sr Ss School	100%

Scholastic Achievements

- Qualified **NSEP 2023** and was among the State Top 1% Nov'22
- Cleared **NDA/NA** written examination **3 times** in a row conducted by **UPSC** . Sept'23, Apr'24, Sept'24

Key Projects

Samsung Smartphone Market Repositioning | Course Project MBA631 | Prof. Amit Shukla (June'25 - July'25)

Objective	• Reassess Samsung’s smartphone market positioning and identify opportunities to strengthen consumer relevance and emotional differentiation.
Approach	• Conducted comprehensive market and competitive analysis of Samsung smartphones across product portfolio, pricing, distribution, and promotions. • Executed primary consumer research (60+ survey responses) to diagnose perception gaps between spec-led messaging and lifestyle expectations. • Synthesized insights to design a repositioning strategy focused on usability, design, trust, and everyday relevance. • Translated strategy into execution by conceptualizing and producing a consumer-facing advertisement aligned with the new positioning.
Impact	• Reframed Samsung smartphones from a spec-centric offering to a lifestyle-driven, emotionally resonant brand proposition.

GNN-Based Fraud Detection System | Electrical Engineering Association IITK (Dec'25 - March'26)

Objective	• Learn and apply Machine Learning (ML), Artificial Neural Networks (ANNs), and Graph Neural Networks (GNNs) to detect illicit transactions using relational patterns in financial networks.
Approach	• Modeled transactions as a graph (nodes: transactions, edges: flows) to capture multi-hop dependencies. • Implemented GCN, GraphSAGE, and GAT for semi-supervised node classification on a 200K+ node dataset. • Addressed class imbalance using Focal Loss and compared performance with an XGBoost baseline.
Impact	• Achieved Macro F1 ≈ 0.92 and PR-AUC ≈ 0.92 , enabling interpretable fraud detection via GNNExplainer.

Signals to Software | Course project EE200 | Prof. Tushar Sandhan | 🌐 (June'26 - July'26)

Objective	• Developed an end-to-end signal processing framework for image restoration, ECG arrhythmia detection, and audio fingerprinting using frequency-domain and time-frequency analysis.
Approach	• Applied Fourier Transform, STFT, notch filtering, and Sobel edge detection for image enhancement and feature extraction. • Implemented beat-wise ECG arrhythmia detection using normalized correlation and spectrogram-based analysis. • Built and deployed a Shazam-inspired audio fingerprinting system using spectrogram peak detection, hash generation, and offset-based matching.
Impact	• Achieved robust multimedia and biomedical signal recognition through scalable DSP pipelines and an interactive Streamlit application.

DocuMind – AI Document RAG Platform | Association of Computer activities (ACA) IITK | 🌐 (June'26 - Ongoing)

Objective	• Built an enterprise-grade Retrieval-Augmented Generation (RAG) platform enabling secure, real-time conversational search and knowledge extraction from large document repositories.
Approach	• Developed a full-stack RAG architecture using Next.js, FastAPI, PostgreSQL, and pgvector. • Implemented hybrid retrieval combining vector embeddings and BM25 for accurate search. • Implemented SSE-based response streaming, Google OAuth authentication, JWT authorization, and tenant-level data isolation for secure multi-user deployment.
Impact	• Enables accurate, secure, and scalable document intelligence by reducing information retrieval time and delivering context-aware answers from large unstructured knowledge bases.

Technical Skills

- **Programming Languages:** C, C++, Python, L^AT_EX, HTML, CSS
- **Software and Libraries:** Git, Github, Numpy, Arduino IDE, Tinker CAD

Relevant Courses (*:Ongoing)

Fundamentals of Computing	Introduction to Probability Theory	Set Theory and Logic
Liner Algebra	Abstract Algebra	Financial Economics
Introduction to Management	Marketing Management	Introduction to Electronics

Positions of Responsibility

Senator BT/BS Y24
| *Students’ Senate , IIT Kanpur*

(April’25-Present)

- Elected by **1200+** undergraduate students (BT/BS Y24) to represent the batch in the Students’ Senate 2025–26.
- Nominated as the Students’ **Senate Nominee to CoSHA** (Council of Students for Hostel Affairs), acting as a liaison between the Senate and CoSHA and supporting the Convener, CoSHA in executive and coordination tasks.
- Created a comprehensive **HEC Guide**, documenting roles, procedures, and best practices to support future Hall Executive Committees across all halls of residence.
- Oversaw scrap collection across all halls of residence, coordinating logistics to enable a centralized scrap auction.

Secretary
| *Robotics Club IITK*

(July’25-Present)

- Core member of a 24-member team, contributing to robotics projects, competition preparation, and club operations.
- Represented Robotics Club IIT Kanpur at the Technoxian World Robotics Championship in the Line Follower category.
- Organized and conducted technical workshops and orientation lectures for freshers to build foundational robotics skills.
- Mentored multiple student project teams, guiding design, implementation, debugging, and successful project completion.
- Supported end-to-end project execution, ensuring technical quality and adherence to timelines.

Secretary
| *President’s Office | Students’ Gymkhana, IIT Kanpur*

(June’25-Present)

- Assisted the President, Students’ Gymkhana in executing administrative, coordination, and campus-wide student initiatives.
- Oversaw and supported the campus-wide auction of lost bicycles, coordinating logistics and processes, generating 4 lakh+ revenue for the Gymkhana budget.
- Contributed to organizing Open House sessions for JEE Advanced qualifiers and Gymkhana orientation for incoming undergraduate students.
- Assisted in Diwali-related student engagement initiatives, including collection and distribution of Diwali wishes to faculty and event coordination.
- Played an active role in organizing Gymkhana Day 2024–25, marking the completion of the Gymkhana’s tenure, including event execution and networking dinner arrangements.

Secretary
| *Hospitality & Transport | Antaragni, IIT Kanpur*

(July’25-Present)

- Managed databases of colleges and contingents, coordinating invitations and participation for Antaragni IIT Kanpur.
- Coordinated with external companies to secure **sponsorships** for participant hospitality kits.
- Oversaw hospitality and accommodation planning for **4000+** participants, coordinating room allocation and logistics across **15 hostels**.
- Assisted in managing transport logistics, including coordination of arrivals, departures, and intra-campus movement of contingents.

Extra-Curriculars

Mentorship	• Mentored 15 freshmen through <i>RoboBreak</i> and <i>RoboDive 2.0</i> delivering structured robotics training and hands-on guidance culminating in functional robot prototypes
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